

BONES VIBRATION

AI Music Creator

Prompt Control Room

User guide • Version 0.22.0

Dense Suno-oriented prompts, local track and image analyzers, EBU R128 loudness metering, and studio WAV export. Runs in the browser or as a Windows desktop app.

AI Music Creator - Prompt Control Room

Version 0.22.0

A Next.js app for building dense, reproducible prompts for AI music workflows (especially Suno-like layouts): genres, grooves, sounds, lyric direction, presets, optional reference analyzers, and export blocks that respect Style / Lyrics field limits. Ships as a static web app, an optional Electron Windows installer, and a Tauri desktop build with native DSP export and Python sidecar integration.

Highlights (v0.22.0)

- Vocal align in project bundle - export/import carries align-preview JSON with instrumental/guide names.
- MusicGen sketch badges - track analyzer shows MusicGen + melody/highlight mode chips.
- Highlight merge e2e - Playwright asserts ·HL in merged Suno rules after highlight generate.
- Maestro coach scroll - step coach can scroll to Maestro chat (data-testid="maestro-chat-panel").
- Co-Producer LLM advisory - sketch brief includes melody/highlight mode context.

Highlights (v0.21.0)

- MG merge tags - Suno AUDIO rule line suffixes ·mel / ·HL for melody and highlight-melody MusicGen sketches.
- Vocal align persistence - alignment preview cached per instrumental; survives panel navigation until track changes.
- Align & export handoff e2e - Playwright coverage for one-click align + handoff pack.
- Step coach scroll - Vocal Embed coach actions scroll to the studio panel.
- Genre model label - track analyzer shows sidecar HF genre model when DistilHuBERT (or override) is active.

Highlights (v0.20.0)

- MusicGen highlight melody e2e - Playwright path: patch highlight range !' melody + highlight checkboxes !' merge (skips without generate_available).
- Maestro highlight melody - "Regenerate highlight melody" chip/command with useHighlightMelody artifact; Co-Producer notes highlight sketches.
- Vocal Embed align & export handoff - one-click align JSON + plan/audio pack for external vocal stacks.
- E2E analysis patch hook - aimc-e2e-patch-audio-analysis for highlight-range tests in dev.

Highlights (v0.19.0)

- Vocal Embed align e2e - Playwright path for alignment preview + one-click align & synthesize (skips without sidecar).
- ci-vocal-ml workflow - optional GitHub Actions job for vocal DSP e2e and vocal-ml stack smoke.
- Maestro LLM native artifacts - system prompt requires model-populated style/lyrics/hooks; melody command enriches musicGenPrompt.
- sidecar:vocal - lightweight scipy install script for vocal synthesis without full torch stack.

Highlights (v0.18.0)

- Vocal Embed one-click path - Align & synthesize preview; handoff pack includes align-preview.json.
- Maestro melody regenerate - generateMusicGenMelody command + suggestion chip when a MusicGen sketch is loaded.
- Co-Producer hooks + sketch - LLM hooks see loaded MusicGen analyzer context.

- MusicGen highlight melody - optional waveform highlight region as melody reference.
- CC0 multi-tag + surprise - toggle multiple tags; surprise pick from filtered pool; reference blocks in import pool.

Highlights (v0.17.0)

- Maestro LLM full artifacts - stylePrompt, lyrics, and hooks in LLM JSON + offline enrich fallbacks; copy chips match offline Maestro.
- Co-Producer LLM hooks - Generate Hooks uses the same LLM settings when enabled; heuristic fallback on errors.
- MusicGen sketch in advisory - Improve Prompt references loaded MusicGen analyzer reports (prompt, BPM, key, melody mode).
- Melody-conditioning e2e - Playwright path: upload track !' melody mode !' generate & merge (skips without generate_available).

Highlights (v0.16.0)

- Maestro LLM suggestion chips - LLM JSON includes suggestions; fallback chips when omitted; same tap-to-send UX as offline Maestro.
- MusicGen live e2e - Playwright spec skips unless sidecar reports generate_available; optional ci-musicgen.yml workflow.
- CC0 import in CI - awesome-suno-sync job re-imports at rotation offset 0 and fails on catalog drift.

Highlights (v0.15.0)

- Maestro LLM artifacts - musicGenPrompt in LLM JSON responses; auto-filled when generateMusicGen fires; copy chip in chat.
- Co-Producer LLM advisory - Improve Prompt uses the same LLM settings as lyrics when enabled; heuristic fallback on errors.
- MusicGen merge tests - unit coverage for buildAudioAnalyzerPatch MG rule line.
- CC0 rotation - import script rotates the 400-concept cap monthly (+ metal/classical/latin/folk/reggae/k-pop sources); override with AIMC_CC0_ROTATE_OFFSET.

Highlights (v0.14.0)

- MusicGen !' Suno loop - generate & play merges into Suno fields (AUDIO + MG prompt line); librosa re-analyze on generated WAV; melody conditioning from current track (POST /generate/melody).
- WaveSurfer default - pro waveform editor on by default (classic highlight editor opt-out via localStorage).
- Vocal Embed depth - MFA/heuristic alignment preview, handoff pack export, RVC smoke spec (skips unless configured).
- CC0 tag filters - filter 400 community concepts by genre keyword in Style Prompt Library.
- Co-producer / Maestro - MusicGen sketch hints and Maestro musicGenPrompt artifact.

Highlights (v0.13.9)

- MusicGen playback - Generate & play loads WAV into track analyzer waveform; Maestro/co-producer can trigger sketches.
- CC0 library - 400 concepts (cap reached).

Highlights (v0.13.7)

- CC0 concept expansion - awesome-suno import now includes examples/ markdown + era/trend supplemental lines (up to 400 cap).

- MusicGen scaffold - opt-in POST /generate sidecar endpoint (npm run sidecar:generate; CC-BY-NC weights).
- WaveSurfer prototype toggle - one-click "Try WaveSurfer prototype" in the track analyzer (persists in localStorage).

Highlights (v0.13.6)

- Vocal Embed E2E polish - guide vocal + lyrics defaults to lyric synthesis with optional MFA/OpenVPI timing; OpenVPI status in UI; synthesis engine returned from sidecar header.
- Style library quick picks - Metaphor roll, Era anchors, 2026 trends, and CC0 concept shortcuts in the Style prompt library.
- Suno lyric consistency - instrumental Lyrics box matches the [Instrumental] / [No Vocals] scaffold everywhere.

Highlights (v0.13.5)

- Research roadmap - BPM evocative descriptors, v5.5 negative guard packs, era-anchored + 2026 trend catalogs, Maestro metaphor surprise rolls.
- Sidecar upgrades - optional DistilHuBERT genre model (AIMC_GENRE_MODEL), CLIP zero-shot vision tags on /analyze-image, MFA guide-vocal alignment hook for OpenVPI DiffSinger .ds timing.
- Catalog refresh - stayen/suno-reference re-sync; awesome-suno import cap raised to 400.

Highlights (v0.13.4)

- Maestro catalog grounding - local retrieval from style catalog + CC0 concepts for LLM prompts (maestro-catalog-grounding.js).

Highlights (v0.13.3)

- BLIP image captioning - optional POST /analyze-image via sidecar vision extra; merged into pixel analyzer.
- CC0 awesome-suno concepts - npm run import:awesome-suno into Style Prompt Library.

Highlights (v0.13.2)

- OpenVPI DiffSinger - real inference via diffsinger_openvpi.py (variance + acoustic); configure AIMC_DIFFSINGER_ROOT + checkpoint exps.

Highlights (v0.13.1)

- Maestro polish coach - step 6 nudges track analysis, voice character, lyrics, and Vocal Embed preview.
- Vocal Embed E2E smoke - npm run test:smoke:vocal validates plan handoff with sidecar running.
- .env.vocal loader - copy ai-sidecar/.env.vocal.example !' ai-sidecar/.env.vocal for model paths without system env.

Highlights (v0.13.0)

- RVC / DiffSinger integration - configure user-owned models via AIMC_RVC_MODEL, AIMC_RVC_API_URL, or AIMC_DIFFSINGER_CMD/URL; GET /vocal-embed/models + UI stack badges.
- Engines - rvc-conversion-v1, diffsinger-v1, plus existing placement-mix and DSP paths.
- npm run sidecar:vocal-ml - installs torch vocal stack for model integrations.

Highlights (v0.12.2)

- Vocal DSP v1 - sidecar vocal extra enables guide vocal pitch conversion (guide-conversion-v1) and lyrics-only synthesis (lyrics-synth-v1) from section timing; health reports vocal_ml_available.

- Vocal Embed Studio - stack status badges, lyrics-only preview when vocal DSP is on, guide file optional in lyrics mode.

Highlights (v0.12.1)

- Vocal Embed placement-mix v1 - sidecar POST /vocal-embed/synthesize ducks the instrumental under lyric sections and overlays a guide vocal; UI adds guide vocal upload and Synthesize placement-mix preview (downloads WAV). Optional vocal extra adds scipy HPF; future vocal-ml reserved for RVC/DiffSinger.

Highlights (v0.12.0)

- Focused Suno step UI - when Prompt Engine is Suno-like, only panels for the current guided step are shown (Style Presets on step 1, Music Controls on groove step, Analyzers on Polish, etc.). Show all tools restores the full studio; preference persists locally.
- Maestro step coach - after you work on a step, a banner nudges you to Proceed to next step or Apply one-click fixes (genre anchors, lyric draft, Suno validator cleanup).
- Vocal Embed Studio - plan local vocal placement on an analyzed instrumental (section timing, voice character, mix brief) without Suno's generation engine. Export JSON or Send plan to sidecar (POST /vocal-embed/plan) for validation and future DiffSinger/RVC synthesis.
- High-precision Voice Character - richer trait map (pitch range, vibrato, jitter/shimmer, articulation) for mimic-style Suno voice prompts.
- Audio upgrades - FLAC uploads with librosa sidecar fallback when the browser cannot decode; browser MP3 export raised to 256 kbps; WAV 16/24-bit and native LAME MP3 via dsp-core unchanged.

Highlights (v0.11.0)

- Maestro workspace commands - chat can now run the app: "use the track analysis" / "use the image analysis" merges analyzer DNA into Suno fields, "take me to polish" / "final step" jumps the guided path. Works offline and via the LLM backend (sanitized command whitelist).
- Remix & extend in chat - "remix it" rerolls groove/palette/mood while keeping your genre identity; "extend the track" lengthens the structure (bridge !' final chorus !' outro) to match Suno's Extend workflow.
- LLM provider presets - one-click OpenAI, OpenRouter, Groq, Mistral, Ollama (local), and LM Studio (local) setup in Co-Producer settings; local providers work without an API key. The same backend powers Maestro Chat.
- Analyzer-aware suggestions - when a track or image analysis exists, Maestro proactively suggests merging it.

Highlights (v0.10.0)

- Maestro - AI Chat Music Creator - a conversational co-producer panel: describe the track ("dark techno at 140 bpm with whispered vocals") and Maestro sets genres, tempo, mood sliders, vocals, sounds, and lyric direction in the project as you talk. Ask for lyrics, hooks, the "d1000-char Suno style prompt, or "surprise me". Fully offline heuristic engine with an optional OpenAI-compatible LLM mode (reuses Co-Producer settings); all LLM patches are sanitized to a key whitelist before touching project state. Chat history persists locally (60 turns).
- Performance & dead-code pass - loudness re-measurement now keys on the audio source (no full blob re-decode on unrelated edits); removed redundant memoized prompts and unused exports across audio-enhancer, export formats, studio export client, and video handoff.
- Dependency refresh - Next, ESLint config, Tailwind, Vitest, Electron 43, and friends bumped to latest patch/minor releases.
- 24 new unit tests for the Maestro engine (parsing, replies, patch sanitizing, LLM protocol).

Highlights (v0.9.13)

- Tauri studio release - GitHub Actions builds Windows/macOS/Linux installers; tag studio-v* to publish (studio-v0.9.13).
- Demucs stem separation - Track analyzer can call the sidecar /separate endpoint and download stem WAVs when the stems extra is installed.
- Native MP3 export (Tauri) - dsp-core encodes MP3 via LAME; Tauri studio export accepts "mp3" alongside WAV.
- Video Creator handoff (Tauri) - Parity with Electron for opening exported audio in the video workflow.
- Version sync - npm run sync:version aligns Tauri, dsp-core, sidecar, package-lock.json, APP_VERSION fallback, and MusicBrainz User-Agent from package.json.

Highlights (v0.9.11)

- Style-DNA Search - Search by artist/title, Spotify track URL, or YouTube link; optional Spotify audio features (danceability, energy, valence, tempo, key) map to Suno genres, mood, and paste-ready style tokens. MusicBrainz fallback when Spotify is not configured.
- Apply Style DNA - One-click merge into project fields with REF: rule line, Goal reference, and guided Polish step.
- 148 unit + 41 e2e tests - Style-DNA mapper unit tests and MusicBrainz-mocked e2e smoke.

Highlights (v0.9.10)

- Shareable project bundles - Export Bundle JSON includes project snapshot, custom style presets, character voice presets, and studio session; import merges presets and still accepts legacy flat project files.
- Suno re-import loop - Paste finished Suno Style/Lyrics, compare vs project-built paste, use pasted fields for Prompt Preview/copy, or apply pasted Lyrics back into generated lyrics.
- 144 unit tests - project bundle parse/export and Suno re-import diff/override coverage.
- 39 e2e tests - bundle export/import smoke and Suno re-import panel flow.

Highlights (v0.9.9)

- Standard engine preview cleanup - Prompt Preview and Copy Prompt use the same paste-ready Style/Lyrics slices as Suno-like (comma-separated Style, bracket Lyrics scaffolds) instead of STYLE: / RULES: blocks.
- Paste-ready e2e smoke - Playwright asserts Prompt Preview, Lyric field preview, and clipboard copy contain no DNA:, theme:, sounds:, or section-label wrappers.
- 136 unit + 36 e2e tests - four new paste-ready preview specs plus existing suite.

Highlights (v0.9.8)

- Paste-ready prompt boxes - Style, Lyrics, analyzer extract, voice style, and guided-path copy blocks output Suno-ready text only (comma-separated Style tokens, bracket Lyrics scaffolds; no DNA: / theme: / tips in <pre> or copy payloads).
- Voice Character Studio integrations - Guided Polish step hints, Co-Producer voice context in lyrics/hooks, Analyze vocal character !' handoff from the track analyzer.
- Online Suno catalog sync - npm run sync:suno-catalog refreshes meta-tags, structure tags, and trademark guards from stayen/suno-reference into the language index and validation rules.
- 136 unit + 32 e2e tests - usable Style/Lyrics export, analyzer style prompt, catalog sync, and studio handoff coverage.

Highlights (v0.9.7)

- YouTube URL sync on session restore - linked reference URL shows in the studio input after import, reload, and preset load (draft state only while typing before link).

- Reset clears voice studio session - Reset to Default clears active analysis, YouTube ref, and compact lines; saved character presets are kept (like custom style presets).
- 123 unit + 31 e2e tests - session reload YouTube assertion and Reset-to-Default studio coverage.

Highlights (v0.9.6)

- Character preset undo snapshots - revert restores characterVoicePresets; capture before studio import/delete.
- Studio session in project JSON - optional characterVoiceStudioSession (analysis, compact Style/metatag, YouTube ref, preset name) on export, autosave, import, and undo.
- 123 unit + 30 e2e tests - session reload and snapshot revert coverage for Voice Character Studio.

Highlights (v0.9.5)

- Character presets in project JSON - optional characterVoicePresets on export, manual save, and autosave; restored on import and reload.
- 118 unit + 28 e2e tests - project import with bundled character presets survives page reload.

Highlights (v0.9.4)

- Voice Character Studio hardening - normalize presets from localStorage; clear stale YouTube refs on load; safe textureTags validation on import.
- Studio UX polish - compact Style line + lyric metatag copy blocks; YouTube URL field syncs on preset load/clear; hardened preset import (functional merge, read errors, storage checks on regenerate).
- 116 unit + 27 e2e tests - adds YouTube-clear and compact-metatag Playwright coverage for Voice Character Studio.

Highlights (v0.9.3)

- Voice Character Studio - analyze vocal files for trait-based Suno prompt DNA (register, breath, vibrato, dynamics); optional YouTube metadata link; save/load character presets.
- Character preset JSON - export and import character presets as JSON (envelope or bare map); merge into localStorage.
- 115 unit + 25 e2e tests - Voice Character Studio Playwright coverage (import, load, export, regenerate, vocal analyze).

Highlights (v0.9.2)

- Snapshot field extraction - pickSnapshotFields + useSnapshotFields slim the workspace provider; snapshot keys live in one place in project-state.js.
- Guided Suno e2e - Playwright walks factory preset !' step navigation !' final Style copy under the 1000-character cap (18 e2e tests total).
- 103 unit tests - adds coverage for pickSnapshotFields alongside the existing suite.

Highlights (v0.9.1)

- Thin page shell - useProjectWorkspaceProvider orchestrates hooks; page.jsx is ~61 lines (layout only).
- Modular workspace - pipeline input, workspace value builder, and snapshot helpers split into dedicated modules.
- E2e workflow coverage - 17 Playwright tests: audio/image analyzer merge !' Suno copy, instrumental !' timed lyrics, plus shared test helpers.
- 101 unit tests - unchanged green suite with release CI running full npm run check.

Highlights (v0.9.0)

- Architecture refactor - useReducer project state, persistence/actions hooks, three-column workspace layout, and co-producer engine extracted from the main page (~780 lines vs ~2,035).
- Reliability - route error boundary, 60s Co-Producer LLM abort/timeout, safeLocalStorage, pinned deps, release CI runs full npm run check.
- Performance - memoized workspace context, React.memo on heavy panels, prompt pipeline keyed on analyzer summary only (not waveform peak edits).
- Tests - 101 unit tests (audio cache + IndexedDB, co-producer engine, audio analyzer) plus Playwright e2e smoke coverage.

Highlights (v0.8.4)

- Reset clears Lyric Style Prompt - blank placeholder until vocal mode and theme are set; generated lyrics/hooks cleared on reset.
- E2e + unit tests for lyric prompt blank slate after Reset to Default.

Highlights (v0.8.3)

- Analyzer catalog sync - audio/image DNA suggestions map to Suno genre/instrument/rhythm pills (BPM-aware).
- 8 new factory presets - Trap, Pop, Lo-Fi, Reggaeton, Metal, Jazz, Afrobeats, Detroit Techno.
- Expanded vocal roles - whispered, raspy, autotuned, duet, crowd chant, stacked harmonies.
- Co-Producer LLM - Spanish/non-English section-tag rules in system prompts.
- Lyric prompt cleanup - language rules without redundant boilerplate.

Highlights (v0.8.2)

- Suno lyric languages - full language catalog (strong + extended tiers), grouped Language selector, Suno section-tag rules in lyric prompts and Co-Producer.
- Suno genres & instruments - ~130 genres, ~150 instruments/textures, searchable grouped pickers; Style Prompt Library still adds 900+ genre-wheel fusion phrases.
- CI e2e - Playwright smoke tests run on every push/PR alongside unit tests and build.

Highlights (v0.8.1)

- Reset to Default !' blank slate - clears all preselected genres, sounds, rules, and lyrics so you can build step by step from guided step 1 (factory presets stay in the sidebar).
- Reset e2e test - Playwright coverage for blank-slate reset behavior.

Highlights (v0.8.0)

- Co-Producer Generate Lyrics - after picking Lyric Style, generate editable lyric drafts matched to that style prompt (Another take, stale-style warning, char budget).
- Style-aware hooks - hook sketches tied to the same Lyric Style presets.
- Instrumental !' lyrics - timed scaffold plus Co-Producer singable draft and hooks in one step.
- Optional LLM backend - OpenAI-compatible API (key stored locally); falls back to built-in templates.
- Swedish / mixed language phrase pools per style.

Highlights (v0.7.9)

- Studio export fix - Electron desktop no longer stalls at 5%; falls back to main-thread export when Web Workers are unavailable on file://.

Highlights (v0.7.8)

- Lyrics on instrumentals - after uploading a track, Add lyrics timed to this track builds a [Verse]/[Chorus] scaffold locked to BPM, duration, and highlight/drop.
- Auto vocal switch - leaves Instrumental mode, merges track DNA, strips “no vocals” rules, and jumps to the Lyric direction step.

Highlights (v0.7.7)

- Cleanup tooling - npm run cleanup:dist:sudo and npm run cleanup:sudo for locked electron-dist* folders on Windows.
- Version fallback sync - in-app version matches package.json when env injection is unavailable.
- Export tests - WAV 24-bit filename and PCM header coverage in vitest.

Highlights (v0.7.6)

- WAV 24-bit export - honest third studio format alongside 16-bit WAV and MP3.
- Undo keeps compact waveforms - peak arrays "d4096 samples restore on revert.
- Desktop update controls - “Check for updates” and “Restart to install” in the packaged app header.
- Pinned Electron toolchain - electron@41.3.0 and electron-builder@26.8.1 for reproducible builds.

Highlights (v0.7.5)

- Undo snapshot fix - capture uses the correct slim state (guided step, variations, history).
- Honest studio export - WAV and MP3 only; legacy flac requests map to WAV.
- Release workflow - push a v* tag to publish the Windows installer and latest.yml for auto-update.
- Vitest 4 - clears the npm audit advisory on the test runner.

Highlights (v0.7.4)

- Turbopack root - turbopack.root in next.config.js fixes dev crashes when Next mis-detects app/ as the workspace.
- Dev debug - npm run debug uses a single inspector on port 9241 (scripts/debug-dev.ps1); npm run stop also clears stray Next node processes.
- Undo - Snapshot restores guided step, variations, and prompt history; audio rehydrates from IndexedDB when cached.
- MP3 fallback - If MP3 encode fails in the worker, export falls back to WAV with a status message.
- npm run cleanup:dist - Removes locked electron-dist* folders when Windows/Cursor releases them.

Highlights (v0.7.2)

- Analyzer honesty - In-app banners clarify that track/image scans are local heuristics, not ML classification.
- Lyrics priority trim - Suno Lyrics paste matches Style: theme/language/sections first, long bodies trim from the end ("d5000).
- Studio export - Background worker with progress bar; WAV / MP3; highlight-loop export.
- Undo snapshot - Revert to last snapshot before preset load, import, merge, or variations.
- Variation A/B - Side-by-side compare with changed-line summary.
- CI - GitHub Actions runs npm test + npm run check on push/PR.
- Electron auto-update - Checks GitHub releases when running a packaged build.

Highlights (v0.7.1)

- Guided Suno path - Step-through workflow, Polish step, progressive style preview, Style capped at 1000

characters with priority ordering on copy.

- Expanded English style vocabulary - Large curated catalog including world/regional styles paired with instruments, sound-design FX, environment beds, orchestral and band instruments, moods, and fusion labels; English-only picker with dedupe.
- Track analyzer (local) - Drop WAV / MP3 / OGG / M4A for a Sonoteller-style report: editable summary, genres, moods, instruments, BPM/key estimates, and a highlight section with full-track + zoomed waveforms (drag amber handles to set the range).
- Audio DNA !' Suno - Merge into Suno fields applies tempo, genres, sounds, rhythms, mood sliders, a compact AUDIO: rule line, and copies the track summary into Goal and Notes when present.
- Image analyzer - Drop JPG / PNG for palette-driven genre/sound suggestions; merges into IMAGE: rule lines and guided fields.
- LUFS meter (EBU R128) - After attach/analyze, shows gated integrated LUFS and true peak (dBTP) using a BS.1770-4-style engine (libebur128-aligned K-weighting and oversampling).
- Studio WAV export - Three local mastering presets: Streaming (" 14 LUFS integrated + " 1 dBTP limit), Wide spatial (stereo width), Punch (low-end and dynamics). Exports 16-bit stereo WAV in the browser.
- Waveform persistence - Autosave/history omit heavy peak arrays; IndexedDB caches audio for rehydrate. Export JSON keeps full waveformPeaks; import preserves them. Attach audio reconnects playback when the cache is missing.
- Version-aware reset - A major package.json version bump clears the autosaved project and analyzer state; presets and history are kept. Patch/minor bumps migrate the saved project in place.
- Refactored UI - Analyzer logic in use-analyzers, splash/header in app-shell, splash timing via useSyncExternalStore (no hydration mismatch in dev).
- Packaged assets - public/bones-logo.webp, root icon.ico, and build/AI_Music_Creator_README.pdf included for Electron builds.
- Live length readout - Style box and lyrics direction character counts next to analyzers (same strings as the validator).
- Presets & history - Factory and custom presets, project save/import/export JSON, variation engine, Co Producer helpers, Suno language index and symbol guides.

Requirements

- Node.js 18+ recommended
- npm (ships with Node)

Development

```
npm install
npm run dev
```

Open <http://localhost:3000>.

Debug (Next.js + Node inspector)

```
npm run debug
```

Starts next dev --inspect=9241 (one inspector port). In VS Code / Cursor, attach to 9241 (see .vscode/launch.json).

Stop stuck dev/debug ports on Windows:

```
npm run stop
```

Remove stale packaging folders (when not locked):

```
npm run cleanup:dist
```

If folders stay locked (Cursor/Electron holding files), run elevated cleanup - it removes what it can and schedules the rest for delete on next reboot:

```
npm run cleanup:dist:admin
```

Old installer output under dist/ is gitignored; safe to delete locally anytime.

Electron auto-update

Packaged builds check GitHub Releases for Druttzen/ai-music-tool on startup. Updates require a published release with latest.yml from electron-builder --publish (or manual upload). Dev/npm run electron skips update checks.

Production build

```
npm run build
```

Pre-ship check (ESLint, zero warnings, then production build):

```
npm run check
```

npm run lint runs ESLint alone.

Static export output is written to out/ (see next.config.js - assetPrefix: "." for Electron-friendly relative assets).

Desktop (Tauri)

Requires Rust (cargo), Tauri CLI, and a Python 3.11–3.12 venv for the sidecar (see ai-sidecar/README.md). On first run, ensure-sidecar-binary.ps1 builds or copies the sidecar binary.

```
npm run tauri:dev
```

Development with sidecar + Tauri webview (same Next dev server as Electron path).

```
npm run tauri:build
```

Production installers under src-tauri/target/release/bundle/. CI publishes tagged releases as **studio-v* (separate from Electron v* tags). Every push to master also runs a tauri-smoke** compile check (Linux cargo build + sidecar bundle).

Sidecar + Demucs stems

See ai-sidecar/README.md. Quick path:

```
npm run bootstrap      # Rust + Python 3.12 if missing (Windows)
npm run sidecar:stems  # one-time torch + demucs install (~2 GB)
npm run sidecar
npm run test:smoke:stems # Demucs UI e2e (slow on CPU)
```

Desktop (Electron)

```
npm run dist
```

Runs npm run build, regenerates build/AI_Music_Creator_README.pdf from this README (npm run build:readme-pdf), prepares out/ for Electron (npm run prepare:electron-dist), then electron-builder. Installer output under electron-dist/ (see package.json build section). The PDF opens once on first launch (main.js).

Regenerate the PDF alone:

```
npm run build:readme-pdf
```

If electron-dist/win-unpacked is locked, prepare:electron-dist automatically falls back to electron-dist-fresh, electron-dist-v{version} (e.g. electron-dist-v071), or a timestamped electron-dist-build-* folder. Close any running AI Music Creator instance and Explorer windows in old output folders if you want the default electron-dist/ path back.

Saved data

Storage | What

localStorage | Autosaved project (ai_music_creator_visual_tool_v3), custom presets, prompt history (slim snapshots - no waveform peaks in history/autosave)

sessionStorage | Splash intro seen flag

IndexedDB (ai-music-creator) | Cached audio blobs for waveform rehydrate (not in exported JSON)

Export JSON | Full project including audioAnalysis.waveformPeaks; re-import restores peaks and timeline (attach audio on another machine for playback if cache is empty)

Studio export notes

- Processing runs locally in the browser; long tracks can take time (max 10 minutes per export).
- Streaming targets " 14 LUFS integrated loudness - common for Spotify/YouTube-style delivery, not a substitute for a certified broadcast meter or mastering engineer.
- Wide spatial is stereo enhancement, not Dolby Atmos.

Version source of truth

The UI reads APP_VERSION from package.json via NEXT_PUBLIC_APP_VERSION in next.config.js. Bump package.json version for releases, then run npm run sync:version so Tauri, dsp-core, sidecar, lockfile, and JS fallbacks stay aligned.

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